



THE 4-STAGE VLM VISUAL & SPATIAL AFFORDANCE GROUNDING PIPELINE

Lifting Open-Vocabulary Natural Language to Metric $SE(3)$ Contact Frames on Edge Accelerators

STAGE 1

Semantic Attention

Multimodal Transformer Ingest

Input: "Pick up the heated bolt"

Vision: RGB-D Patch Tokens

Compute: Cross-Attention Matrix

Substrate: Linux MPU / NPU

Aligns language tokens with visual feature maps

STAGE 2

Visual Grounding

2D Pixel Bounding & Mask

Mask: Centroid $(u_c, v_c) \in \mathbb{R}^2$

Extent: Bounding box $[u_1, v_1, u_2, v_2]$

Confidence: $p_{\text{sem}} \geq \gamma_{\text{thresh}}$

Resolution: 640×480 Depth Map

Isolates candidate object pixels from background

STAGE 3

3D De-Projection

Metric Ray Casting (K^{-1})

Model: $p_{\text{cam}} = Z \cdot K^{-1}[u, v, 1]^T$

Bounds: $\mathcal{B} \subset \mathbb{R}^3$ Metric Box

World: $p_{\text{world}} = T_{\text{cam}}^{\text{world}} \cdot p_{\text{cam}}$

Centroid: $[x_0, y_0, z_0]^T \in \mathbb{R}^3$

Lifts 2D pixel coordinates into calibrated 3D space

STAGE 4

Affordance Frame

$SE(3)$ Grasp Lease Target

Pose: $T_{\text{target}}^{\text{world}} \in SE(3)$

Approach: Normal vector $\mathbf{n}_{\text{app}}$

Aperture: Gripper width d_{grasp}

Force Limit: $F_{\text{contact}} \leq F_{\text{max}}$

Emits complete metric intent lease $\mathcal{L}_{\text{intent}}$



OUTPUT CONTRACT: Expiring Intent Lease $\mathcal{L}_{\text{intent}} = \langle T_{\text{target}}^{\text{world}} \in SE(3), \mathcal{B} \subset \mathbb{R}^3, \dot{\mathbf{x}}_{\text{max}} \leq 0.40 \text{ m/s}, t_{\text{issue}}, \Delta t_{\text{TTL}} = 800 \text{ ms} \rangle$